Deep Residual Learning for Image Recognition

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Abstract

Deeper neural networks are more difficult to train. We present a residual learning framework to ease the training of networks that are substantially deeper than those used previously. We explicitly reformulate the layers as learning residual functions with reference to the layer inputs, instead of learning unreferenced functions. We provide comprehensive empirical evidence showing that these residual networks are easier to optimize, and can gain accuracy from considerably increased depth. On ImageNet, we evaluate residual nets with a depth of up to 152 layers—8x deeper than VGG nets but still having lower complexity.

1. Degradation Problem

When deeper networks are able to start converging, a degradation problem has been exposed: with network depth increasing, accuracy gets saturated and then degrades rapidly. Unexpectedly, such degradation is not caused by overfitting, and adding more layers to a suitably deep model leads to higher training error, as verified by extensive experiments on CIFAR-10 and ImageNet.

Residual Mapping Formulation

Section 3: Deep Residual Learning Formulation and Shortcut Connections

3.1 Residual Building Block

Let us consider H(x) as an underlying mapping to be fit by a few stacked layers, with x denoting the inputs to the first of these layers. We hypothesize that multiple nonlinear layers can asymptotically approximate residual functions H(x) - x. Rather than expecting stacked layers to approximate H(x), we explicitly let these layers approximate a residual mapping F(x) = H(x) - x. The original mapping is recast into:

$$y = F(x, \{W_i\}) + x$$

3.2 Identity Shortcut Connections

The operation F + x by shortcut connections and element-wise addition introduces neither extra parameters nor computation complexity. The dimensions of x and F must be equal. When changing input/output channels, linear projection W_s can be performed: y = F(x, {W_i}) + W_s x.

ImageNet Benchmarks & Bottleneck Architecture

Section 5: Empirical Benchmarks on ImageNet 2012 Classification

5.1 ImageNet Error Rates Table

ImageNet classification top-1 and top-5 error rates for plain networks vs. residual networks:

Network	Layers	FLOPs	Top-1 Error	Top-5 Error
VGG-16	16	15.3 x 10^9	28.07%	9.33%
VGG-19	19	19.6 x 10^9	28.05%	9.12%
ResNet-34	34	3.6 x 10^9	21.84%	5.71%
ResNet-50	50	3.8 x 10^9	20.74%	5.25%
ResNet-101	101	7.6 x 10^9	19.87%	4.60%
ResNet-152	152	11.3 x 10^9	19.38%	4.49%

6. Bottleneck Design and Conclusions

For deeper networks such as ResNet-50, 101, and 152, we use a 3-layer bottleneck block consisting of 1x1, 3x3, and 1x1 convolutions. The 1x1 layers are responsible for reducing and then restoring dimensions, leaving the 3x3 layer a bottleneck with smaller input/output dimensions.